Tech Bite #9: Measuring SWR and the Antenna System

Your antenna and feed line are a system. This Bite is about checking that it works -- mainly with one number, SWR.

Group T7C · 11 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

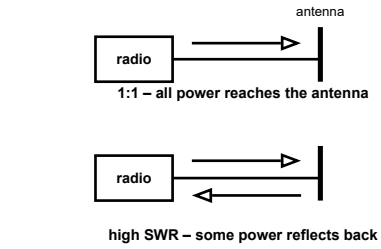
The Big Picture

Your antenna and the feed line connecting it to the radio are a **system**. This Bite is about measuring whether that system is working – and most of it comes down to one number: **SWR**.

SWR: The Match Between Radio and Antenna

SWR – standing wave ratio – tells you how well the antenna is matched to the feed line. A reading of **1:1** is a perfect match: all the power goes out. (T7C04) A reading like **4:1** means an **impedance mismatch** – power is reflected back instead of radiated. (T7C06)

Because high SWR can harm the radio, most solid-state transmitters automatically **reduce power as SWR rises**, to protect their output transistors. (T7C05) You can read SWR with a **directional wattmeter**. (T7C08)

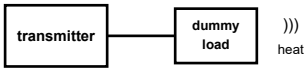


SWR is the match. 1:1 sends all power out; high SWR reflects power back toward the radio.

Testing Safely: The Dummy Load

When you want to test a transmitter **without putting a signal on the air**, you use a **dummy load**. (T7C01) A dummy load is a **50-ohm non-inductive resistor mounted on a heat sink** – it absorbs the transmitter's power as heat instead of radiating it. (T7C03)

To check whether an antenna is **resonant** at your operating frequency, use an **antenna analyzer**. (T7C02)



Power becomes heat – no signal goes on the air. Safe for testing.

A dummy load absorbs the transmitter's power as heat, so you can test without transmitting on the air.

The Feed Line

Power lost in a feed line is converted into **heat**. (T7C07) Coaxial cable fails mainly from **moisture getting in** (T7C09), which is why the outer jacket must resist **ultraviolet light** – UV damages the jacket and lets water enter. (T7C10) And **foam-dielectric** coax has **less loss per foot** than solid-dielectric coax. (T7C11)

Mental Cheat Sheet

- **SWR 1:1** = perfect match, all power radiated.
- **SWR 4:1** = an impedance mismatch.
- Solid-state radios **cut power** as SWR rises, to protect output transistors.
- Read SWR with a **directional wattmeter**.
- A **dummy load** = 50-ohm non-inductive resistor on a heat sink; test without going on the air.
- An **antenna analyzer** checks antenna resonance.
- Feed-line loss becomes **heat**.
- Coax fails from **moisture**; foam dielectric has **less loss**.

Common Traps

- A **perfect** match is **1:1** – not "zero" and not "50:50."
- 4:1 is a **mismatch** – it is not a gain or a loss in dB.
- Radios cut power at high SWR to protect **transistors** – not for FCC rules.
- A dummy load is a **resistor** – not a power supply and relay.
- Coax fails mostly from **moisture**, not from high duty cycle.
- Feed-line loss turns into **heat**, it does not become harmonics.

Takeaway: Your antenna system is judged by **SWR** – 1:1 is perfect, higher numbers mean a mismatch with power reflected back, and solid-state radios protect themselves by cutting power when SWR climbs. Use a **dummy load** to test without transmitting on the air, and an **antenna analyzer** to check resonance. Keep moisture out of your coax and its loss low.

Bite 9 Question Bank (T7C)

T7C01. What is the primary purpose of a dummy load?
A. To prevent transmitting signals over the air when making tests
B. To prevent over-modulation of a transmitter
C. To improve the efficiency of an antenna
D. To improve the signal-to-noise ratio of a receiver

T7C02. Which of the following is used to determine if an antenna is resonant at the desired operating frequency?
A. A VTVM
B. An antenna analyzer
C. A Q meter
D. A frequency counter

T7C03. What does a typical RF dummy load consist of?
A. A low-voltage power supply and an AC relay
B. A 50-ohm non-inductive resistor mounted on a heat sink
C. A low-voltage power supply and a DC relay
D. A 50-ohm inductive reactance mounted in a shielded enclosure

T7C04. What reading on an SWR meter indicates a perfect impedance match between the antenna and the feed line?
A. 50:50
B. Zero
C. 1:1
D. Full Scale

T7C05. Why do most solid-state transmitters reduce output power as SWR increases beyond a certain level?
A. To protect the RF output amplifier transistors
B. To comply with FCC rules on spectral purity
C. Because power supplies cannot supply enough current at high SWR
D. To lower the SWR on the transmission line

T7C06. What does an SWR reading of 4:1 indicate?
A. Loss of −4 dB
B. Good impedance match
C. Gain of +4 dB
D. Impedance mismatch

T7C07. What happens to power lost in a feed line?
A. It increases the SWR
B. It is radiated as harmonics
C. It is converted into heat
D. It distorts the signal

T7C08. Which instrument can be used to determine SWR?
A. Voltmeter
B. Ohmmeter
C. Iambic pentameter
D. Directional wattmeter

T7C09. Which of the following causes failure of coaxial cables?
A. Moisture contamination
B. Solder flux contamination
C. Rapid fluctuation in transmitter output power
D. Operation at 100% duty cycle for an extended period

T7C10. Why should the outer jacket of coaxial cable be resistant to ultraviolet light?
A. Ultraviolet light can increase the resistance of the conductors
B. Ultraviolet light can increase losses in the cable’s jacket
C. Ultraviolet and RF signals can mix, causing interference
D. Ultraviolet light can damage the jacket and allow water to enter the cable

T7C11. What is an advantage of foam-dielectric versus solid-dielectric coaxial cable?
A. It is more resistant to moisture contamination
B. It has higher voltage breakdown
C. It has less loss per foot
D. It has a better impedance match to 50 ohms

Answer Key

T7C01: A T7C02: B T7C03: B T7C04: C

T7C05: A T7C06: D T7C07: C T7C08: D

T7C09: A T7C10: D T7C11: C